Online Supplement to Hard to Beat: Historical Rules versus Conditional Quantile Models for Peak Net-Position Limits

This compact supplement contains the existing mathematical proofs and the complete principal adverse-result inventories, with links to unchanged longer archives. It accompanies the v7 manuscript and does not add models, data, statistical tests or parameter selection. All later diagnostics are identified as post-result / exploratory; the two original primary comparisons retain their values and roles.

Tables format saved native fields. Loss percentages are the existing relative differences against the named reference; probabilities are labelled as such. Different source-native monetary units are not pooled. NA and failed rows remain distinct from complete results. The exact binding file and document builder permit source-field readback without re-running any financial model.

The complete earlier supplement, full comparison families, all candidate tables, annual and sparse sensitivities, numerical details and source guides remain accessible through EXTENDED RESULTS. This index is an access route, not a substitute for compatible input data or independent validation.

S1 Known-floor projection and strict quantile coverage

We forecast the sampled maximum of the absolute position, including the observed issue position. Write $B = |N_t|$ and let $Y \geq B$ almost surely, with B measurable at the forecast origin. For $0 < \tau < 1$, the pinball score is $\rho_\tau(u) = \tau u_+ + (1-\tau)(-u)_+$. A lower inverse conditional quantile is $q_\tau = \inf\{c : F(c \mid \mathcal{F}_t) \geq \tau\}$.

Proposition 1: floor projection. Replacing a report c by $\max(B, c)$ weakly reduces its score at every possible outcome. At the population conditional quantile, the strict breach probability is at most $1 - \tau$. These are distinct statements: projection applies to any report; the tail qualification requires the true conditional quantile.

$$\rho_\tau(Y - c) - \rho_\tau(Y - \max\{B, c\}) = \tau(B - c)_+, \qquad \Pr(Y > q \mid \mathcal{F}_t) \leq \alpha.$$

Proof. If $c \geq B$, the replacement changes nothing. If $c < B$, both errors $Y - c$ and $Y - B$ are nonnegative, so their score difference is $\tau(B - c)$. This proves the pointwise identity without a density assumption. Right continuity of the conditional distribution gives $F(q_\tau \mid \mathcal{F}_t) \geq \tau$, hence $\Pr(Y > q_\tau \mid \mathcal{F}_t) \leq 1 - \tau$. An atom at the quantile can make this strict; exact nominal hit frequency is not required for a valid upper quantile. A fitted report does not automatically inherit this population claim. *QED*

Why the floor matters. The sampled maximum equals the current magnitude whenever the recorded future path remains inside its current absolute-position band. This can create an atom

even when later signed observations are continuous. The existence of a floor alone does not imply an atom: an outcome uniformly distributed above the floor provides a counterexample. Equality-based coverage tests therefore diagnose their stated equalities, rather than every possible valid upper-quantile report.

Endpoint marginals do not identify the peak. The exact artificial constructions in the formal example record hold each future-time marginal fixed while changing temporal dependence. Comonotone and disjoint rare events then give different peak quantiles. The example is a logical construction, not another financial dataset or an estimated mechanism. Forecast evaluation must use the ordered path response and the information available at its actual origin.

Takeaway. Known-floor projection is an elementary pointwise improvement. It does not establish empirical calibration, identify the conditional distribution, or turn a fitted endpoint forecast into a forecast of a path maximum.

S2 Classical endpoint-to-peak bound

Proposition 2: the classical endpoint bound. Let future real increments $X_1, \dots, X_H$ be independent and individually symmetric about zero. Set $S_0 = 0$, $S_h = \sum_{j=1}^{h} X_j$, and fix a starting position x. Equal increment laws and finite moments are unnecessary. For the population absolute-displacement quantile at level $1-\alpha/2$, $q = \inf\{c : \Pr(|S_H| \le c) \ge 1-\alpha/2\}$, the classical scalar maximal inequality and the triangle inequality give

$$\Pr\left(\max_{0\le h\le H} |x + S_h| > B + q\right) \le \Pr\left(\max_{h} |S_h| > q\right) \le 2\Pr(|S_H| > q) \le \alpha.$$

This is a direct application of the classical Lévy inequality, not a new maximal inequality. The existing source audit identifies Sussman (2023), Proposition 3.4, and records the actual consulted source range.

Proof, including strict events and atoms. For $c \ge 0$, let η be the first index with $|S_\eta| > c$. On $\{\eta = k, S_k > c\}$, the remaining sum $R_k = S_H - S_k$ is independent of the information through k and is symmetric. Therefore $\Pr(R_k \ge 0) \ge 1/2$, including a possible atom at zero, and that weak sign keeps $S_H > c$. On the corresponding negative crossing, $R_k \le 0$ keeps $S_H < -c$, also with probability at least one half. Summing over the disjoint crossing events gives $\Pr(\max_h |S_h| > c) \le 2\Pr(|S_H| > c)$. The argument covers $c = 0$. Finally, $\max_{0\le h\le H} |x + S_h| \le |x| + \max_h |S_h|$, and right continuity gives the required strict tail bound at q. *QED*

Conditional use and empirical implementation. For an observed random anchor, the same argument must hold under its forecast-origin conditional law, with the corresponding conditional population endpoint quantile. Conditional independence and zero symmetry cannot be inferred from unconditional marginals. The empirical endpoint plug-in adds the current floor to a historical absolute-displacement order statistic. Overlapping labels, estimated quantiles and changing dependent histories provide no automatic inheritance of this population bound or a pinball-risk ordering.

Two different failure mechanisms. Dependence can destroy the inequality: take $X_1 = Z$, $X_2 = -Z$, with $Z \in \{-1, 1\}$ equally likely. The endpoint is zero, while the sampled maximum

is one. Separately, replacing the required absolute-displacement quantile by a generic one-sided endpoint quantile can fail even for independent symmetric increments. The exact finite example is preserved in the formal example record. These logical counterexamples are separate from the Gaussian Monte Carlo cases below; a nonviolating simulation does not prove a theorem extension.

Takeaway. The bound supplies a clear population benchmark under explicit assumptions; the empirical report remains a model to validate, not a coverage certificate.

S3 Monte Carlo mechanisms: original increment persistence

These are the existing fixed-seed Gaussian experiments, not newly run simulations or calibrated replicas of either financial source. Every reported rate is a probability, not a percentage. The peak includes the initial level; the endpoint event concerns absolute displacement from that initial level. Each path cell has 100000 paths. L/Stat starts the level AR in stationarity; L/Abs fixes its absolute-level tail anchor; L/Sig is the separate signed-level tail-anchor sensitivity. I/Zero accumulates stationary AR increments from a zero position. M denotes a stationary-marginal endpoint threshold; C denotes a distinct fixed-start conditional endpoint oracle. Threshold partners sharing a seed use the same paths, not independent repetitions.

Table S1: Original increment-AR simulation, unchanged. Serial persistence here concerns increments, not fitted position levels.

H	ϕ	Process	Seed	q	Hits	Peak prob.	MC SE	End prob.
6	0.00	I/Zero	20267023	5.490	3035	0.03035	0.00054	0.02454
6	0.70	I/Zero	20267093	10.092	2597	0.02597	0.00050	0.02573
6	0.85	I/Zero	20267108	11.634	2550	0.02550	0.00050	0.02550
6	0.98	I/Zero	20267121	13.190	2557	0.02557	0.00050	0.02557
24	0.00	I/Zero	20285023	10.981	3819	0.03819	0.00061	0.02504
24	0.70	I/Zero	20285093	24.599	2772	0.02772	0.00052	0.02475
24	0.85	I/Zero	20285108	33.394	2619	0.02619	0.00051	0.02525
24	0.98	I/Zero	20285121	49.784	2572	0.02572	0.00050	0.02572

The independent-increment rows satisfy the classical premise. Positive increment coefficients violate independence without necessarily causing a numerical breach of the bound. Their favourable rates are not a general result for serial dependence.

S4 Mean-reverting levels and negative-increment sensitivities

The corrected level process is $N_h = \phi N_{h-1} + \varepsilon_h$, with independent Gaussian innovations of variance $1-\phi^2$ and unit stationary level variance. Its stationary increments have lag correlation $\rho_X(k) = -(1-\phi)\phi^{k-1}/2$, so this mechanism differs from a positively persistent increment AR. The fitted financial coefficient describes levels, making the level equation the relevant illustrative mechanism. The negative-increment family is a separate sensitivity, with stationary increments and nonstationary cumulative levels.

Table S2: All corrected primary cases. Mean-reverting level rates are conservative in these finite cases; the negative-increment result must also be retained.

H	ϕ	Process	Seed	q	Hits	Peak prob.	MC SE	End prob.
6	0.68	L/Stat/M	20567123	3.009	175	0.00175	0.00013	0.02462
6	0.68	L/Abs/M	20567124	3.009	0	0.00000	0.00000	0.10447
6	0.85	L/Stat/M	20567133	2.502	300	0.00300	0.00017	0.02553

H	ϕ	Process	Seed	q	Hits	Peak prob.	MC SE	End prob.
6	0.85	L/Abs/M	20567134	2.502	14	0.00014	0.00004	0.08227
6	0.98	L/Stat/M	20567143	1.071	913	0.00913	0.00030	0.02434
6	0.98	L/Abs/M	20567144	1.071	372	0.00372	0.00019	0.03623
6	-0.30	I/Zero/M	20567223	4.244	3563	0.03563	0.00059	0.02524
6	-0.60	I/Zero/M	20567233	3.127	4439	0.04439	0.00065	0.02558
24	0.68	L/Stat/M	20585123	3.170	636	0.00636	0.00025	0.02495
24	0.68	L/Abs/M	20585124	3.170	1	0.00001	0.00001	0.11416
24	0.85	L/Stat/M	20585133	3.138	402	0.00402	0.00020	0.02520
24	0.85	L/Abs/M	20585134	3.138	1	0.00001	0.00001	0.11159
24	0.98	L/Stat/M	20585143	1.965	768	0.00768	0.00028	0.02468
24	0.98	L/Abs/M	20585144	1.965	85	0.00085	0.00009	0.06201
24	-0.30	I/Zero/M	20585223	8.167	4281	0.04281	0.00064	0.02476
24	-0.60	I/Zero/M	20585233	5.701	5032	0.05032	0.00069	0.02516

All saved level cases have peak rates below the nominal threshold. The largest negative-increment point estimate occurs at the longer horizon and the more negative coefficient. Its rate is 0.05032, with a saved excess of 0.032 percentage points and Monte Carlo SE 0.00069. This is not demonstrated population failure. General dependence invalidates the classical premise regardless of the observed direction. Zero observed counts also do not establish zero population probability.

S5 All secondary simulation sensitivities

For stationary random starts, absolute endpoint displacement has variance $2(1-\phi^H)$. At a fixed start x, its conditional Gaussian mean and variance are $(\phi^H - 1)x$ and $1 - \phi^{2H}$. Thus holding the stationary-marginal endpoint threshold across starts does not condition on a large anchor. The conditional-oracle sensitivities below are distinct thresholds on shared paths; neither is an oracle for the maximum.

Table S3: All secondary anchor and conditional-threshold sensitivities. Conditional and marginal endpoint calibration remain distinct.

H	ϕ	Process	Seed	q	Hits	Peak prob.	MC SE	End prob.
6	0.68	L/Abs/C	20567124	3.717	0	0.00000	0.00000	0.02411
6	0.68	L/Sig/M	20567125	3.009	2	0.00002	0.00001	0.06261
6	0.68	L/Sig/C	20567125	3.433	1	0.00001	0.00001	0.02476
6	0.85	L/Abs/C	20567134	3.036	0	0.00000	0.00000	0.02488
6	0.85	L/Sig/M	20567135	2.502	21	0.00021	0.00005	0.05689
6	0.85	L/Sig/C	20567135	2.840	4	0.00004	0.00002	0.02607
6	0.98	L/Abs/C	20567144	1.146	215	0.00215	0.00015	0.02503
6	0.98	L/Sig/M	20567145	1.071	464	0.00464	0.00021	0.03184
6	0.98	L/Sig/C	20567145	1.117	340	0.00340	0.00018	0.02521
24	0.68	L/Abs/C	20585124	3.920	0	0.00000	0.00000	0.02567
24	0.68	L/Sig/M	20585125	3.170	9	0.00009	0.00003	0.06486
24	0.68	L/Sig/C	20585125	3.605	1	0.00001	0.00001	0.02514
24	0.85	L/Abs/C	20585134	3.880	0	0.00000	0.00000	0.02495
24	0.85	L/Sig/M	20585135	3.138	3	0.00003	0.00002	0.06306
24	0.85	L/Sig/C	20585135	3.571	0	0.00000	0.00000	0.02484
24	0.98	L/Abs/C	20585144	2.298	20	0.00020	0.00004	0.02518
24	0.98	L/Sig/M	20585145	1.965	146	0.00146	0.00012	0.04607
24	0.98	L/Sig/C	20585145	2.179	56	0.00056	0.00007	0.02556

The complete correction contains 34 threshold rows on 22 initialized path samples. The retained CSV and portable readback provide exact endpoint moments, deviations from the nominal probability, and the single-prespecified-row bound for zero counts. None estimates financial innovation laws, changing fitted parameters or empirical-quantile uncertainty.

S6 Empirical endpoint plug-in: all original cells

The endpoint plug-in uses each completed historical episode's absolute endpoint displacement and adds its specified empirical quantile to the actual current-position floor. It shares the historical rule's completed origin pool and fixed refit schedule, with no fitted tuning parameter. The operational empirical rank and overlapping financial history are distinct from the known population quantile in the classical proposition. The report therefore has to earn its financial coverage and score performance empirically.

Table S4: Every original-cell endpoint plug-in comparison. A population bound does not imply a lower empirical pinball score.

Source	H/W	L/F Δ%	L/QR Δ%	Hits L/F
FX	6/720	30.93	33.76	101/332
FX	6/2880	30.39	52.98	75/360
FX	24/720	10.85	-15.71	136/427
FX	24/2880	7.94	12.16	174/357
Bills	6/52	3.37	-43.62	10/11
Bills	6/260	-11.77	-41.49	15/28
Bills	24/52	26.32	-61.42	42/33
Bills	24/260	9.87	-47.64	22/12

Table S5: All separate endpoint-plug-in working-reference contrasts. Adverse score comparisons remain alongside lower-breach reports.

Source	H/W	Ref.	DM t	DM p	HLN p	Holm p
FX	6/720	F	7.891	2.988e-15	3.658e-15	4.483e-14
FX	6/720	QR	6.608	3.908e-11	4.37e-11	5.081e-10
FX	6/2880	F	7.790	6.695e-15	8.123e-15	9.373e-14
FX	6/2880	QR	12.832	1.089e-37	3.451e-37	1.743e-36
FX	24/720	F	1.848	0.06462	0.06562	0.5169
FX	24/720	QR	-1.673	0.09434	0.09556	0.6604
FX	24/2880	F	2.058	0.03958	0.04032	0.3958
FX	24/2880	QR	1.530	0.1259	0.1273	0.7002
Bills	6/52	F	0.971	0.3317	0.3422	0.7002
Bills	6/52	QR	-2.778	0.005471	0.006887	0.06018
Bills	6/260	F	-1.345	0.1785	0.1884	0.7002
Bills	6/260	QR	-3.031	0.002441	0.003236	0.02929
Bills	24/52	F	1.569	0.1167	0.1522	0.7002
Bills	24/52	QR	-1.991	0.04645	0.06948	0.418
Bills	24/260	F	1.092	0.2746	0.3183	0.7002
Bills	24/260	QR	-1.516	0.1294	0.1664	0.7002

The family and its doubled-bandwidth sensitivities are separate from the original regression family. The complete tables keep every adverse direction; no negative comparison is removed by describing the population inequality as conservative. The linked records preserve source-native mean error amounts and every numerical status.

S7 AR residual replay and its adverse comparisons

At each original refit, estimate an intercept-bearing level recursion on each unique mature adjacent transition once. Replaying a historical ordered residual block from the current position

gives the exact algebraic identity

$$M_{i,h}(x) = N_{i+h} + \widehat{\phi}^{\,h}(x - N_i), \qquad M_{i,0}(x) = x.$$

The empirical peak uses the same completed scenario origins, response rank, floor and refit schedule as its matched historical controls. Finite negative or above-unit coefficients were eligible, not clipped; all observed fitted coefficients in the archived run happened to lie inside the unit interval. Their descriptive means are not identified data-generating parameters.

Table S6: Complete original AR replay comparison. Lower scores can accompany more strict breaches.

Source	H	W	Mean $\widehat{\phi}$	AR/S Δ%	AR/QR Δ%	Hits AR/S
FX	6	720	0.981	-1.54	2.45	282/255
FX	6	2880	0.985	-0.55	16.20	338/268
FX	24	720	0.981	2.39	-25.32	401/289
FX	24	2880	0.985	5.15	8.68	444/292
Bills	6	52	0.676	-13.44	-52.66	22/11
Bills	6	260	0.847	-18.89	-47.21	31/28
Bills	24	52	0.711	35.13	-59.43	56/33
Bills	24	260	0.850	-24.12	-63.86	59/12

The replay improves on original regression in every original Treasury cell but loses in most FX cells. Its comparisons with the translated-path rule have mixed directions, while strict breaches increase in every row above. Sparse-grid and annual signs can reverse. The full paired, annual and fixed-grid tables remain in the AR archive. These are post-result / exploratory same-data calculations, not independent validation. Numerical readback checked the represented-input recursion and saved summaries without validating the financial innovation law.

S8 Ordinary quantile ranks and minimizer endpoints

The ordinary empirical quantile and the corrected empirical rule use the same completed pool, refit times and issue floor. The former takes the lower empirical pinball minimizer; the latter uses the specified add-one rank. At integer probability-times-pool-size values, adjacent reports can be the two endpoints of the same empirical minimizer interval. Calling the lower report a uniquely optimal comparator would therefore be wrong. At the short weekly pool, the noninteger rank distinction differs.

Table S7: Complete ordinary-rank ablation. Lower empirical reports increase breach occasions and need not improve score.

Source	H	W	QR/I Δ%	I/F Δ%	Hits I/F/QR
FX	6	720	-1.967	-0.153	337/332/531
FX	6	2880	-14.773	0.008	365/360/376
FX	24	720	31.508	-0.003	432/427/924
FX	24	2880	-3.783	0.026	359/357/531
Bills	6	52	87.086	-1.995	14/11/84
Bills	6	260	46.058	3.237	32/28/54
Bills	24	52	208.258	6.213	37/33/132
Bills	24	260	105.020	2.353	15/12/84

The reconstructed historical summaries match the original controls. The lower report increases strict hits and mean underprediction amount in every cell, but its loss direction is mixed. Every Treasury regression remains worse than the lower empirical report. This ablation isolates a rank convention on the same viewed record; it neither supplies a fresh validation sample nor establishes universal superiority of either finite-sample quantile endpoint. Exact counts, annual splits and fixed-grid rows are linked in the complete rank archive.

S9 Fixed-report error preferences

For a fixed report, changing the loss weight traces a line $R_a = aU + (1-a)V$, where $U = \text{mean}(P-C)_+$ and $V = \text{mean}(C-P)_+$. These are passive error amounts over all origins, not observed funding costs. Breach frequency averages an indicator and need not rank reports like either amount. The following source-native monetary values are never pooled across FX and weekly aggregates.

Table S8: The four fixed headline comparisons. The longer-horizon FX result is sensitive to error weights.

Comparison	U_L	U_R	V_L	V_R	Equality a
fx_h6_primary	156123.03	166531.06	4750560.18	5671390.59	none
fx_h6_matched	156123.03	185672.23	4750560.18	5526087.27	none
fx_h24_legacy	317678.12	149813.88	8658017.18	12121206.97	$\frac{86579744869851975}{90776350660766843}$
treasury_h6_primary	3131.86	1395.64	23688.09	28656.17	$\frac{24840389716843550}{33521504184965797}$

The equality column gives exact saved rational boundaries, not newly estimated costs. The short-horizon FX primary reduces both mean error amounts; its elementary-score curves can nevertheless cross because a Murphy diagram varies the elementary score threshold, a different comparison from reweighting these two unconditional amounts. The longer-horizon FX legacy pair reverses near the loss-weight ratio reported in the main text. Actual decision costs were not observed.

The complete frontiers below include every open region and every boundary tie. The full and sparse samples remain separate, and feasible reports outside the specified menu are not part of these menu-relative envelopes. All pairwise relations, dominated labels and exact arithmetic are additionally preserved in the preference archive.

S10 Complete preference frontier: full sample

Open intervals exclude their endpoints. Point rows retain all ties, including ties at the extreme loss weights. Method codes here denote the original fixed report menu; their definitions remain in the main methods table. The sparse grid is a sensitivity on the same record, not independent data.

Table S9: Complete saved preference frontier, full sample. Exact ties prevent an artificially unique winner.

Task	Kind	Left / point a	Right a	Minimizers
FX full H6/W720	open	$\frac{0}{1}$	$\frac{105760946788801400}{110990640317093439}$	QR
	open	$\frac{105760946788801400}{110990640317093439}$	$\frac{19056123729472220}{19757413338275541}$	KNN
	open	$\frac{19056123729472220}{19757413338275541}$	$\frac{1}{1}$	frozen_S
	point	$\frac{0}{1}$		QR
	point	$\frac{105760946788801400}{110990640317093439}$		KNN, QR
	point	$\frac{19056123729472220}{19757413338275541}$		KNN, frozen_S

Task	Kind	Left / point a	Right a	Minimizers
FX full H6/W2880	point	$\frac{1}{1}$		frozen_S
	open	$\frac{0}{1}$	$\frac{34631329313121430}{35599009063461303}$	QR
	open	$\frac{34631329313121430}{35599009063461303}$	$\frac{1}{1}$	SCALED_KNN
	point	$\frac{0}{1}$		QR
	point	$\frac{34631329313121430}{35599009063461303}$		QR, SCALED_KNN
	point	$\frac{1}{1}$		SCALED_KNN
FX full H24/W720	open	$\frac{0}{1}$	$\frac{32241742214205920}{38133893639725449}$	QR
	open	$\frac{32241742214205920}{38133893639725449}$	$\frac{9205010760823860 0}{103514765282707447}$	KNN
	open	$\frac{9205010760823860 0}{103514765282707447}$	$\frac{290754316347088 60}{30985073840047543}$	AR1_REPLAY
	open	$\frac{290754316347088 60}{30985073840047543}$	$\frac{1}{1}$	frozen_S
	point	$\frac{0}{1}$		QR
	point	$\frac{32241742214205920}{38133893639725449}$		KNN, QR
	point	$\frac{9205010760823860 0}{103514765282707447}$		AR1_REPLAY, KNN
	point	$\frac{290754316347088 60}{30985073840047543}$		AR1_REPLAY, frozen_S
	point	$\frac{1}{1}$		frozen_S
FX full H24/W2880	open	$\frac{0}{1}$	$\frac{335769696945034900}{350844274558596259}$	QR
	open	$\frac{335769696945034900}{350844274558596259}$	$\frac{1}{1}$	frozen_S
	point	$\frac{0}{1}$		QR
	point	$\frac{335769696945034900}{350844274558596259}$		QR, frozen_S
	point	$\frac{1}{1}$		frozen_S
Bills full H6/W52	open	$\frac{0}{1}$	$\frac{3320065684462664}{4624714213835193}$	QR
	open	$\frac{3320065684462664}{4624714213835193}$	$\frac{1563059024367757 00}{160668673434229883}$	AR1_REPLAY
	open	$\frac{1563059024367757 00}{160668673434229883}$	$\frac{1}{1}$	KNN, frozen_F
	point	$\frac{0}{1}$		QR
	point	$\frac{3320065684462664}{4624714213835193}$		AR1_REPLAY, QR
	point	$\frac{1563059024367757 00}{160668673434229883}$		AR1_REPLAY, KNN, frozen_F
	point	$\frac{1}{1}$		KNN, frozen_F
Bills full H6/W260	open	$\frac{0}{1}$	$\frac{8679397304452847}{9039550431967810}$	AR1_REPLAY
	open	$\frac{8679397304452847}{9039550431967810}$	$\frac{1}{1}$	frozen_K
	point	$\frac{0}{1}$		AR1_REPLAY
	point	$\frac{8679397304452847}{9039550431967810}$		AR1_REPLAY, frozen_K
	point	$\frac{1}{1}$		frozen_K
Bills full H24/W52	open	$\frac{0}{1}$	$\frac{5759715075028296}{135868932262782 37}$	QR
	open	$\frac{5759715075028296}{135868932262782 37}$	$\frac{1176111552346570 25}{130987328519855581}$	frozen_K
	open	$\frac{1176111552346570 25}{130987328519855581}$	$\frac{1}{1}$	frozen_S
	point	$\frac{0}{1}$		QR
	point	$\frac{5759715075028296}{135868932262782 37}$		QR, frozen_K
	point	$\frac{1176111552346570 25}{130987328519855581}$		frozen_K, frozen_S
	point	$\frac{1}{1}$		frozen_S
Bills full H24/W260	open	$\frac{0}{1}$	$\frac{3813577886766790}{4899469998019859}$	frozen_K
	open	$\frac{3813577886766790}{4899469998019859}$	$\frac{9133367261160288 5}{9415677626848826 3}$	AR1_REPLAY
	open	$\frac{9133367261160288 5}{9415677626848826 3}$	$\frac{2408270758122742 50}{2418754512635378 7}$	KNN
	open	$\frac{2408270758122742 50}{2418754512635378 7}$	$\frac{1}{1}$	frozen_S
	point	$\frac{0}{1}$		frozen_K
	point	$\frac{3813577886766790}{4899469998019859}$		AR1_REPLAY, frozen_K
	point	$\frac{9133367261160288 5}{9415677626848826 3}$		AR1_REPLAY, KNN
	point	$\frac{2408270758122742 50}{2418754512635378 7}$		KNN, frozen_S
	point	$\frac{1}{1}$		frozen_F, frozen_S

S11 Complete preference frontier: sparse sensitivity

Open intervals exclude their endpoints. Point rows retain all ties, including ties at the extreme loss weights. Method codes here denote the original fixed report menu; their definitions remain in the main methods table. The sparse grid is a sensitivity on the same record, not independent data.

Table S10: Complete saved preference frontier, fixed sparse grid. Exact ties prevent an artificially unique winner.

Task	Kind	Left / point a	Right a	Minimizers
FX sparse H6/W720	open	$\frac{0}{1}$	$\frac{35228646474289300}{37058573227618587}$	QR
	open	$\frac{35228646474289300}{37058573227618587}$	$\frac{128015277909259100}{130046193358911709}$	KNN
	open	$\frac{128015277909259100}{130046193358911709}$	$\frac{1}{1}$	SCALED_KNN
	point	$\frac{0}{1}$		QR
	point	$\frac{35228646474289300}{37058573227618587}$		KNN, QR
	point	$\frac{128015277909259100}{130046193358911709}$		KNN, SCALED_KNN
	point	$\frac{1}{1}$		SCALED_KNN
FX sparse H6/W2880	open	$\frac{0}{1}$	$\frac{59571292631784500}{60902063609752461}$	QR
	open	$\frac{59571292631784500}{60902063609752461}$	$\frac{1}{1}$	SCALED_KNN
	point	$\frac{0}{1}$		QR
	point	$\frac{59571292631784500}{60902063609752461}$		QR, SCALED_KNN
	point	$\frac{1}{1}$		SCALED_KNN
FX sparse H24/W720	open	$\frac{0}{1}$	$\frac{7530906001239450}{9350542730727077}$	QR
	open	$\frac{7530906001239450}{9350542730727077}$	$\frac{18652943561940400}{20490003118958463}$	KNN
	open	$\frac{18652943561940400}{20490003118958463}$	$\frac{11935316416561250}{12439911923115111}$	AR1_REPLAY
	open	$\frac{11935316416561250}{12439911923115111}$	$\frac{1}{1}$	frozen_S
	point	$\frac{0}{1}$		QR
	point	$\frac{7530906001239450}{9350542730727077}$		KNN, QR
	point	$\frac{18652943561940400}{20490003118958463}$		AR1_REPLAY, KNN
	point	$\frac{11935316416561250}{12439911923115111}$		AR1_REPLAY, frozen_S
	point	$\frac{1}{1}$		frozen_S
FX sparse H24/W2880	open	$\frac{0}{1}$	$\frac{578937865306180}{598942665753459}$	QR
	open	$\frac{578937865306180}{598942665753459}$	$\frac{1}{1}$	frozen_S
	point	$\frac{0}{1}$		QR
	point	$\frac{578937865306180}{598942665753459}$		QR, frozen_S
	point	$\frac{1}{1}$		frozen_S
Bills sparse H6/W52	open	$\frac{0}{1}$	$\frac{1315902266754605}{2037515146643951}$	QR
	open	$\frac{1315902266754605}{2037515146643951}$	$\frac{679337}{682434}$	frozen_K
	open	$\frac{679337}{682434}$	$\frac{1}{1}$	KNN, frozen_F
	point	$\frac{0}{1}$		QR
	point	$\frac{1315902266754605}{2037515146643951}$		QR, frozen_K
	point	$\frac{679337}{682434}$		KNN, frozen_F, frozen_K
	point	$\frac{1}{1}$		KNN, frozen_F
Bills sparse H6/W260	open	$\frac{0}{1}$	$\frac{39729652848099850}{42395294149611417}$	AR1_REPLAY
	open	$\frac{39729652848099850}{42395294149611417}$	$\frac{1}{1}$	frozen_K
	point	$\frac{0}{1}$		AR1_REPLAY
	point	$\frac{39729652848099850}{42395294149611417}$		AR1_REPLAY, frozen_K
	point	$\frac{1}{1}$		frozen_K
Bills sparse H24/W52	open	$\frac{0}{1}$	$\frac{8631914270543611}{12443891775419019}$	QR
	open	$\frac{8631914270543611}{12443891775419019}$	$\frac{2529092855837558}{2690148205992001}$	frozen_K
	open	$\frac{2529092855837558}{2690148205992001}$	$\frac{37657795243316428}{38019573842698655}$	AR1_REPLAY
	open	$\frac{37657795243316428}{38019573842698655}$	$\frac{1}{1}$	KNN, frozen_F, frozen_S
	point	$\frac{0}{1}$		QR
	point	$\frac{8631914270543611}{12443891775419019}$		QR, frozen_K
	point	$\frac{2529092855837558}{2690148205992001}$		AR1_REPLAY, frozen_K
	point	$\frac{37657795243316428}{38019573842698655}$		AR1_REPLAY, KNN, frozen_F, frozen_S
	point	$\frac{1}{1}$		KNN, frozen_F, frozen_S
Bills sparse H24/W260	open	$\frac{0}{1}$	$\frac{19956537365855950}{27652346249496953}$	frozen_K
	open	$\frac{19956537365855950}{27652346249496953}$	$\frac{1}{1}$	AR1_REPLAY
	point	$\frac{0}{1}$		frozen_K

Task	Kind	Left / point a	Right a	Minimizers
	point	$\frac{19956537365855950}{27652346249496953}$		AR1_REPLAY, frozen_K
	point	$\frac{1}{1}$		AR1_REPLAY

S12 Complete weekly panel: definitions and first half

Each cell compares the same peak target, completed-label pool, origin sequence, fixed refit schedule and native monetary units. The original full regression, smaller predictor regression, expanded-ridge full regression and ordinary empirical report all remain visible. The additional series are related weekly aggregates, not independent individual-dealer replications. Changes in predictors and changes in the ridge menu are distinct interventions. Every listed comparison is post-result / exploratory.

The historical column gives mean pinball loss and strict hits. Other columns give saved percentage loss difference against that same historical rule and strict hits; negative percentages favour the column method. Original QR33, QR4 and expanded QR33 are distinct fitted procedures. I is the ordinary empirical-quantile report, with the endpoint qualification in Section S8. Across the archived panel, expanded full-regression median excess loss is 9.22 percent, compared with 92.63 percent for the original menu. Expanded regression still loses in 34 cells. This improvement does not erase the remaining failures or identify a causal feature effect.

Table S11: Complete panel, first half. Additional regularization leaves substantial adverse cases.

Series	H/W	F loss/hits	QR33 %/hits	QR4 %/hits	QR33+ %/hits	I %/hits
PDPOSMBS-TOT	6/52	921.7 / 21	170.97 / 101	3.24 / 27	5.82 / 28	3.26 / 27
PDPOSMBS-TOT	6/260	945.6 / 26	7.36 / 19	1.17 / 27	7.36 / 19	1.96 / 27
PDPOSMBS-TOT	24/52	1753.7 / 39	215.55 / 145	5.09 / 43	66.24 / 104	3.49 / 42
PDPOSMBS-TOT	24/260	1611.9 / 25	26.79 / 29	1.04 / 26	0.72 / 24	1.14 / 26
PDPOSCS-TOT	6/52	530.3 / 16	52.75 / 48	-5.79 / 23	-4.70 / 23	-6.28 / 23
PDPOSCS-TOT	6/260	399.7 / 6	1.90 / 15	-1.10 / 6	-0.95 / 8	-1.56 / 7
PDPOSCS-TOT	24/52	579.4 / 16	73.78 / 63	1.85 / 24	2.97 / 25	2.12 / 23
PDPOSCS-TOT	24/260	569.9 / 6	27.46 / 3	-3.38 / 6	-3.06 / 5	-4.68 / 6
PDPOSFGS-TOT	6/52	400.7 / 16	107.22 / 77	-4.15 / 24	-4.29 / 20	-4.98 / 18
PDPOSFGS-TOT	6/260	358.3 / 6	-1.08 / 14	4.88 / 9	-6.04 / 6	-1.54 / 6
PDPOSFGS-TOT	24/52	476.2 / 36	128.67 / 64	13.36 / 37	10.67 / 37	4.52 / 38
PDPOSFGS-TOT	24/260	354.0 / 9	30.02 / 21	72.24 / 26	-0.97 / 10	-1.50 / 9
PDPOSTIPS-L2	6/52	287.3 / 26	217.18 / 125	6.74 / 29	2.64 / 28	1.69 / 26
PDPOSTIPS-L2	6/260	269.3 / 24	66.28 / 45	1.78 / 25	66.28 / 45	1.63 / 25
PDPOSTIPS-L2	24/52	641.0 / 48	307.36 / 181	24.88 / 69	152.16 / 146	1.88 / 49
PDPOSTIPS-L2	24/260	532.2 / 47	226.91 / 147	11.41 / 59	226.91 / 147	1.84 / 51
PDPOSTIPS-G11	6/52	42.3 / 23	196.70 / 118	1.71 / 29	6.68 / 41	-1.13 / 26
PDPOSTIPS-G11	6/260	41.8 / 14	90.31 / 38	63.18 / 29	-1.52 / 13	-2.53 / 14
PDPOSTIPS-G11	24/52	80.8 / 61	218.69 / 123	135.59 / 103	64.63 / 78	-0.01 / 63
PDPOSTIPS-G11	24/260	51.2 / 9	98.59 / 33	37.18 / 30	98.59 / 33	-1.09 / 10
PDPOSTIPS-G2	6/52	181.3 / 16	208.61 / 96	8.58 / 21	35.47 / 41	3.33 / 19
PDPOSTIPS-G2	6/260	179.4 / 19	63.47 / 36	10.78 / 24	63.47 / 36	-0.29 / 19
PDPOSTIPS-G2	24/52	377.9 / 31	277.81 / 131	11.77 / 42	40.47 / 60	-3.63 / 33
PDPOSTIPS-G2	24/260	286.0 / 23	109.33 / 66	120.12 / 58	109.33 / 66	0.29 / 23

S13 Complete weekly panel: second half

The second half follows the identical convention. All original and expanded results remain unchanged. Cross-series dependence, aggregation offsets, native-unit differences and overlapping economic positions preclude treating these cells as independent replications. Forecasts are fitted separately; observations and losses are not pooled into a single financial population.

Table S12: Complete panel, second half. The enlarged predictor menu is not uniformly better than historical rules.

Series	H/W	F loss/hits	QR33 %/hits	QR4 %/hits	QR33+ %/hits	I %/hits
PDPOSTIPS-G6L11	6/52	133.6 / 9	59.30 / 73	-1.65 / 12	-2.52 / 13	-0.79 / 12
PDPOSTIPS-G6L11	6/260	114.9 / 4	-5.72 / 14	-1.11 / 4	-1.17 / 4	-1.00 / 4
PDPOSTIPS-G6L11	24/52	147.0 / 10	76.48 / 92	-0.97 / 16	-2.43 / 17	0.44 / 15
PDPOSTIPS-G6L11	24/260	133.8 / 2	-26.42 / 1	-1.30 / 2	-1.33 / 2	-0.55 / 2
PDPOSGSC-L2	6/52	1232.9 / 15	215.90 / 129	-3.85 / 23	215.90 / 129	-2.12 / 19
PDPOSGSC-L2	6/260	1247.4 / 15	64.40 / 40	9.04 / 20	2.20 / 20	0.61 / 17
PDPOSGSC-L2	24/52	2270.8 / 46	276.55 / 165	182.80 / 96	131.63 / 93	5.02 / 55
PDPOSGSC-L2	24/260	2918.7 / 33	41.60 / 55	-0.19 / 33	0.12 / 32	0.14 / 34
PDPOSGSC-G2L3	6/52	792.4 / 24	121.15 / 93	145.25 / 112	25.51 / 40	3.86 / 28
PDPOSGSC-G2L3	6/260	1004.0 / 28	21.31 / 44	22.62 / 59	21.31 / 44	0.23 / 30
PDPOSGSC-G2L3	24/52	2101.9 / 47	112.95 / 135	129.55 / 151	112.95 / 135	1.78 / 54
PDPOSGSC-G2L3	24/260	2677.0 / 55	11.60 / 64	39.73 / 91	-2.06 / 52	1.93 / 55
PDPOSGSC-G3L6	6/52	764.9 / 12	87.61 / 82	75.44 / 72	10.10 / 29	-4.25 / 15
PDPOSGSC-G3L6	6/260	699.1 / 20	0.79 / 15	1.91 / 21	0.79 / 15	2.27 / 22
PDPOSGSC-G3L6	24/52	977.4 / 38	544.34 / 137	299.12 / 120	19.44 / 47	9.23 / 45
PDPOSGSC-G3L6	24/260	931.0 / 20	15.52 / 37	0.13 / 20	-5.76 / 17	0.56 / 20
PDPOSGSC-G6L7	6/52	542.3 / 16	253.66 / 142	89.00 / 79	48.88 / 63	-1.24 / 24
PDPOSGSC-G6L7	6/260	564.9 / 20	31.61 / 42	71.05 / 49	31.61 / 42	0.43 / 21
PDPOSGSC-G6L7	24/52	993.4 / 43	264.80 / 217	264.77 / 215	264.80 / 217	11.08 / 47
PDPOSGSC-G6L7	24/260	699.0 / 22	146.41 / 77	142.57 / 67	146.41 / 77	0.98 / 23
PDPOSGSC-G7L11	6/52	505.7 / 25	175.62 / 83	15.72 / 34	86.69 / 55	14.85 / 34
PDPOSGSC-G7L11	6/260	506.2 / 25	8.34 / 31	-0.06 / 25	8.34 / 31	0.44 / 26
PDPOSGSC-G7L11	24/52	1653.9 / 36	94.95 / 105	62.40 / 91	-0.51 / 39	2.38 / 36
PDPOSGSC-G7L11	24/260	1002.9 / 32	229.96 / 94	202.78 / 99	229.96 / 94	8.62 / 35

The independent saved-array arithmetic checks and selection-before- evaluation records are linked from the E5 archive. The original panel's entire candidate/status inventory remains in the original panel archive, and the fitted expanded-menu predictions are not substitutes for unavailable methods or independent dealer data.

S14 Complete added-panel endpoint-plug-in comparison

The original added-panel endpoint plug-in was evaluated on the same task grids as the historical and regression reports. Each entry gives the saved loss percentage against the historical rule, followed by strict hits for the endpoint plug-in and historical report. These are all related aggregate tasks, not independent replications or source-pooled monetary losses.

Table S13: Every additional-panel endpoint-plug-in result. Fewer breaches do not guarantee a better score.

Series	H6/W52	H6/W260	H24/W52	H24/W260
PDPOSMBS-TOT	7.17; 11/21	7.33; 5/26	-17.87; 27/39	-3.92; 22/25
PDPOSCS-TOT	18.28; 11/16	17.56; 2/6	-5.06; 11/16	19.63; 4/6
PDPOSFGS-TOT	-4.74; 13/16	16.77; 0/6	-3.72; 37/36	-4.06; 12/9
PDPOSTIPS-L2	-1.89; 12/26	-7.42; 19/24	-34.60; 30/48	-4.52; 46/47
PDPOSTIPS-G11	13.93; 3/23	27.71; 1/14	-16.39; 16/61	27.11; 0/9
PDPOSTIPS-G2	19.45; 4/16	23.00; 4/19	6.36; 27/31	2.09; 20/23
PDPOSTIPS-G6L11	9.67; 5/9	36.27; 0/4	5.03; 4/10	12.63; 0/2

Series	H6/W52	H6/W260	H24/W52	H24/W260
PDPOSGSC-L2	20.65; 6/15	-2.64; 6/15	0.93; 27/46	-22.23; 24/33
PDPOSGSC-G2L3	17.31; 15/24	-14.81; 12/28	22.35; 49/47	-9.35; 40/55
PDPOSGSC-G3L6	20.57; 9/12	3.66; 8/20	-11.37; 20/38	-5.73; 17/20
PDPOSGSC-G6L7	0.95; 12/16	-10.15; 15/20	1.69; 49/43	5.19; 21/22
PDPOSGSC-G7L11	4.92; 11/25	1.05; 12/25	-20.33; 26/36	1.88; 26/32

The complete original panel CSV also preserves every original regression candidate, selection result, native-unit mean error amount and failure status. This table adds no refit or parameter selection and does not extend the independent-increment population guarantee to these financial series.

S15 Earlier-period fixed selection rules

The earlier-period experiment applies the saved fixed selection rules to familiar historical data. The bounded repository audit found no prior evaluation role for these earlier origins; this is still retrospective, not external or prospective validation. R0 denotes the fixed minimum-validation-loss rule. R2 is the predesignated main one-standard-error selection rule, not a result-dependent replacement chosen for its subsequent performance. The rule design precedes its calculations but follows earlier research results.

R1 adds the ordinary empirical quantile to the finite regression candidates. R2 compares each candidate's validation losses with the same best-mean candidate, using the paired-difference Bartlett-HAC standard error. Candidates within that tolerance qualify, and the fixed rule prefers the strongest shrinkage, including the empirical report. This paired HAC tolerance is a time-series adaptation of a one-standard-error preference, not the textbook formula, a significance test or a noninferiority guarantee. It does not account for the whole uncertainty from selecting the best candidate on that segment.

R3 applies the same preference to prespecified earlier rolling folds and the outer validation segment, with origin-equal weighting and maturity gaps. Its HAC pairs use actual source-index distances rather than compressing gaps. Without an eligible earlier fold, the saved design makes it identical to R2; this is not a numerical fallback. Earlier-period results were sealed before the already-inspected-period calculations that could reuse those origins.

Table S14: Every earlier-period R0/R2 comparison. The one-standard-error rule misses the recorded success conditions.

Series	H/W	R0 %	R2 %	Hits R0/R2	Hits F
PDPOSGS-B	6/52	11.22	7.43	10/10	7
PDPOSGS-B	24/52	181.39	181.39	27/27	8
PDPOSMBS-TOT	6/52	-4.11	3.83	7/6	4
PDPOSMBS-TOT	24/52	48.55	26.55	14/13	10
PDPOSCS-TOT	6/52	-11.56	-11.56	0/0	0
PDPOSCS-TOT	24/52	-4.12	1.02	4/4	2
PDPOSFGS-TOT	6/52	-27.32	-27.32	6/6	0
PDPOSFGS-TOT	24/52	-3.87	-3.60	4/3	2
PDPOSTIPS-L2	6/52	-7.06	-7.06	2/2	1
PDPOSTIPS-L2	24/52	-23.33	-1.55	4/1	1
PDPOSTIPS-G11	6/52	-5.55	-5.79	5/5	4
PDPOSTIPS-G11	24/52	5.67	-0.87	15/15	15
PDPOSTIPS-G2	6/52	15.16	15.16	8/8	0
PDPOSTIPS-G2	24/52	30.27	30.27	14/14	10
PDPOSTIPS-G6L11	6/52	283.88	28.58	17/6	0
PDPOSTIPS-G6L11	24/52	-4.03	-2.70	2/1	0
PDPOSGSC-L2	6/52	48.04	16.09	17/15	14

Series	H/W	R0 %	R2 %	Hits R0/R2	Hits F
PDPOSGSC-L2	24/52	40.77	40.77	34/34	30
PDPOSGSC-G2L3	6/52	1.40	11.14	12/13	8
PDPOSGSC-G2L3	24/52	-11.22	15.48	15/20	15
PDPOSGSC-G3L6	6/52	2.63	2.35	11/11	10
PDPOSGSC-G3L6	24/52	15.62	6.92	29/28	26
PDPOSGSC-G6L7	6/52	19.03	19.03	6/6	1
PDPOSGSC-G6L7	24/52	2.05	1.82	20/19	19
PDPOSGSC-G7L11	6/52	19.18	12.77	13/13	10
PDPOSGSC-G7L11	24/52	8.48	2.70	29/29	27
fx_hourly	6/720	2.44	0.62	79/73	63
fx_hourly	6/2880	-16.62	-6.28	20/20	16
fx_hourly	24/720	-4.36	-4.36	47/47	32
fx_hourly	24/2880	-1.42	-1.40	1/1	0

All recorded success criteria are NOT_MET, NOT_MET and NOT_MET, respectively. R2's worst weekly excess is 181.39 percent, and its weekly median is 5.37 percent. The long-history FX gain also fails the prewritten retention criterion. This adverse result remains a robustness finding rather than a contribution claim. Complete inspected-period and all-rule rows remain linked in the R6 archive.

S16 Selection failures and replication boundaries

Required validation-candidate failure blocks the dependent rule; it is neither retried nor replaced by a shorter mean. The complete noncomplete report inventory is below. Every listed row has the saved reason `REQUIRED_VALIDATION_CANDIDATE_FAILED`. Those failures remain part of the evidence even though other rules in the same task can complete.

Table S15: All noncomplete R6 report rows. Failed mandatory candidates remain failures.

Phase	Series	H/W	Rule	Status
retro	fx_hourly	6/720	R3	FAILED
retro	fx_hourly	24/720	R3	FAILED
inspected	fx_hourly	6/720	R3	FAILED
inspected	fx_hourly	24/720	R3	FAILED

The Treasury earlier-period structural comparison uses a shorter history than the original primary. It is explicitly an analogue, not an exact earlier replication of the long-history primary. The source-native comparison and its definition are preserved in the primary-structure table. All original primary values remain unchanged. Reclassifying the analogue as an exact primary replication, or calling familiar retrospective FX data an unseen dealer, would change the scientific claim without adding evidence.

S17 Finite delayed calibration and headroom SAV

The finite delayed ACI-style wrapper adapts past strict hits after their entire peak labels mature. Its score pool contains only completed residual scores and uses the specified finite memory. The base-plus-score quantile is projected onto the known floor. The adaptive alpha state is projected to the current feasible range; the rank-zero convention issues the floor. Gamma was selected

by coverage distance, then pinball loss and the fixed tie rule, a coverage-first selection criterion distinct from original regression fitting. No canonical ACI theorem is imported to this modified procedure.

The headroom SAV method adapts a CAViaR recursion, rather than replicating a return-VaR model. The completed headroom response is normalized by its training-only scale; the recursive report is multiplied back by that same scale and then floor-projected. Its fixed bounded optimizer and starts retain all eligibility and convergence failures; a complete fit is not a proof of a global optimum. Selection state carries across the unissued chronological gap.

Table S16: Complete delayed-wrapper and headroom-SAV inventory. Two long-horizon wrapper cells remain unavailable.

Source	H/W	Method	Mean loss	Hits	Status
FX	6/720	ACI_F	472677.77	331	COMPLETE
FX	6/720	ACI_QR	439672.91	352	COMPLETE
FX	6/720	SAV	449237.37	349	COMPLETE
FX	6/2880	ACI_F	463271.47	328	COMPLETE
FX	6/2880	ACI_QR	393795.06	329	COMPLETE
FX	6/2880	SAV	456696.22	364	COMPLETE
FX	24/720	ACI_F	921402.08	322	COMPLETE
FX	24/720	ACI_QR	1020220.09	393	COMPLETE
FX	24/720	SAV	781586.36	439	COMPLETE
FX	24/2880	ACI_F	791441.18	323	COMPLETE
FX	24/2880	ACI_QR	755982.70	294	COMPLETE
FX	24/2880	SAV	756811.00	333	COMPLETE
Bills	6/52	ACI_F	3818.40	18	COMPLETE
Bills	6/52	ACI_QR	3489.82	29	COMPLETE
Bills	6/52	SAV	2974.32	16	COMPLETE
Bills	6/260	ACI_F	2792.91	7	COMPLETE
Bills	6/260	ACI_QR	3013.16	6	COMPLETE
Bills	6/260	SAV	2892.23	29	COMPLETE
Bills	24/52	ACI_F	4368.10	38	COMPLETE
Bills	24/52	ACI_QR	6250.53	47	COMPLETE
Bills	24/52	SAV	4182.40	35	COMPLETE
Bills	24/260	ACI_F	NA	NA	NA
Bills	24/260	ACI_QR	NA	NA	NA
Bills	24/260	SAV	3100.97	25	COMPLETE

The NA wrapper rows have the saved reason `VALIDATION_INELIGIBLE`; no test is substituted for the absent forecasts. All losses are in each source's native units and must not be pooled.

S18 Wrapper comparisons, uncertainty and missing cells

Comparing wrapped regression with wrapped historical reports is a different question from comparing either wrapper with its unwrapped base. The following table retains every within-cell comparison. A lower coverage distance measures closeness of empirical hits to the target probability, not conditional coverage or lower monetary loss.

Table S17: Every paired wrapper comparison. Calibration adjustment preserves the long-history short-horizon FX score gain but does not produce universal dominance.

Source	H/W	QR/F Δ%	Hits QR/F	QR dist.	F dist.
FX	6/720	-6.98	352/331	0.00313	0.00004
FX	6/2880	-15.00	329/328	0.00034	0.00049
FX	24/720	10.72	393/322	0.00932	0.00140

Source	H/W	QR/F Δ%	Hits QR/F	QR dist.	F dist.
FX	24/2880	-4.48	294/323	0.00562	0.00125
Bills	6/52	-8.61	29/18	0.05469	0.01498
Bills	6/260	7.89	6/7	0.02834	0.02473
Bills	24/52	43.09	47/38	0.11968	0.08718
Bills	24/260	NA	NA	NA	NA

The working-reference comparison inventory contains 56 fixed slots, with 5 raw p values unavailable. The reported Holm bookkeeping keeps those slots; mapping an NA component to a bookkeeping value does not create an observed p value or validate failed reference assumptions. HLN references remain separately identified. Exact update equations, gamma candidates, selection freezes, numerical receipts and all comparisons remain in the v6 replay guide and the complete WP2 inventory. The original primaries are not overwritten by these post-result / exploratory extensions. The upper-horizon NA rows and adverse SAV comparisons remain visible rather than becoming a favourable subset.

S19 Untuned forecast combinations

Fixed report combinations use the matched historical and original regression reports at identical origins. Their arithmetic mean is untuned. The median also requires the original selected plain local-quantile report; neither a smaller predictor model nor the endpoint plug-in substitutes for that missing constituent. Combining floor-respecting reports preserves the floor, but does not guarantee a conditional quantile or a score below the best constituent. Identical reports generate an NA equal-loss p value rather than a manufactured significance result.

Table S18: All original-cell combination comparisons. A report combination can improve one constituent while losing to the other.

Source	H/W	Rule	vs F %	vs QR %	Hits
FX	6/720	Mean	-6.91	-4.90	338
FX	6/720	Median	-4.81	-2.75	325
FX	6/2880	Mean	-10.61	4.87	330
FX	6/2880	Median	-11.17	4.22	327
FX	24/720	Mean	5.50	-19.77	533
FX	24/720	Median	5.00	-20.15	514
FX	24/2880	Mean	-8.94	-5.39	345
FX	24/2880	Median	-3.74	0.01	416
Bills	6/52	Mean	-3.15	-47.18	26
Bills	6/52	Median	0.00	-45.46	11
Bills	6/260	Mean	2.77	-31.84	40
Bills	6/260	Median	-3.98	-36.32	28
Bills	24/52	Mean	18.87	-63.69	54
Bills	24/52	Median	0.00	-69.46	33
Bills	24/260	Mean	-5.32	-54.88	28
Bills	24/260	Median	-11.51	-57.83	14

Table S19: Complete combination counts including the added panel. Missing median constituents remain NA.

Rule	Total	Complete	NA	Below both	Below F	Below QR	Equal F
Mean	56	56	0	14	16	54	0
Median	56	8	48	3	5	6	2

The added panel lacks the specified plain local-quantile constituent, so all its median rows remain unavailable. Mean combinations are complete, but lower loss than regression alone does not imply lower loss than the historical rule. All panel-specific values and every comparison remain in the complete combination record.

S20 Combination Murphy diagrams and comparison families

Elementary-score comparisons vary the score threshold while keeping the quantile level fixed. Exact integer-event signs at every unique breakpoint determine sample crossing before display interpolation. A picture is not used to select favourable thresholds. These matched combination contexts are distinct from the original FX primary with its shorter rolling reference.

Table S20: Every combination Murphy comparison in the original-primary contexts. Crossing prevents threshold-uniform sample dominance.

Source	Left	Right	Ranking	Positive	Negative
FX	MEAN_F_QR	F	CROSSING	217	15628
FX	MEAN_F_QR	QR	CROSSING	13562	2343
FX	MEDIAN_F_QR_KNN	F	CROSSING	909	13572
FX	MEDIAN_F_QR_KNN	QR	CROSSING	10827	2575
Bills	MEAN_F_QR	F	CROSSING	336	291
Bills	MEAN_F_QR	QR	CROSSING	179	454
Bills	MEDIAN_F_QR_KNN	F	CROSSING	262	274
Bills	MEDIAN_F_QR_KNN	QR	CROSSING	147	446

The complete normal-DM comparison family has 224 slots, of which 98 have NA raw p values. The family adjustment is bookkeeping for those saved working references, not error control over the complete historical analysis search. The inspected record, overlapping labels, repeated fitting and related aggregate series remain common to the methods. All original and adverse rows remain in the combination findings. Exact breakpoint signs support only the stated finite-sample score comparison; they do not certify general forecast dominance, optimal funding, or a causal benefit from acting on the limits.

S21 Training-defined position-change states

The states are low, middle and high issue-known RMS position-change states, not price volatility or yield regimes. Cutpoints come only from the training history. Reports and original primary means stay unchanged; the diagnostic divides the already viewed evaluation record without estimating a switching policy. Source-native group differences are not pooled.

Table S21: All three state groups for each original primary. The FX advantage remains negative in every group, while the weekly evidence remains adverse.

Source	State	n	QR minus reference	HAC SE
FX	low	1593	-58761.27	16492.94
FX	middle	1816	-27839.43	16257.70
FX	high	3216	-70387.91	17895.89
Bills	low	9	6696.66	1705.21
Bills	middle	25	5230.59	2608.15
Bills	high	243	810.88	466.48

Table S22: Complete state diagnostic reference family. A joint rejection need not identify a useful state rule.

Source	Diagnostic	Raw p	Holm p	Status
FX	joint_GW	1.46298e-06	5.8519e-06	OK
FX	group_heterogeneity	0.159383	0.159383	OK
Bills	joint_GW	0.0484878	0.0969756	OK
Bills	group_heterogeneity	0.000426772	0.00128032	OK

These finite-instrument and group-mean HAC references require unverified moment, dependence and covariance assumptions. A joint conditional- moment rejection can be driven by an unconditional mean difference; it is not evidence that a state-dependent trading or limit-selection policy works. The source-level reconstruction and all cutpoints remain in the complete state record.

S22 Refitted target-level sensitivities

Changing the target quantile requires fitting the specified target- level procedure; it is different from reweighting the errors of a fixed report. The following saved sensitivities keep the original horizon and reference structure. The weekly primary and matched reference coincide and count once. All unavailable upper-target rows remain NA with `VALIDATION_INELIGIBLE`, rather than being replaced by a lower target or a favourable subset. Absolute loss values across different quantile levels measure different objectives.

Table S23: Complete target-level sensitivity. The lower-target directions recur, but upper-target evidence remains unavailable.

Source	τ	Reference	Mean difference	QR hits	Ref. hits	Status
FX	0.90	primary	-60715.44	686	620	OK
FX	0.90	matched	-67628.17	686	667	OK
Bills	0.90	primary	1175.03	73	55	OK
FX	0.99	primary	NA	NA	NA	NA
FX	0.99	matched	NA	NA	NA	NA
Bills	0.99	primary	NA	NA	NA	NA

The complete family contains 6 slots, with 3 raw p values unavailable. The existing selection freezes and chronological evaluation receipts remain linked in the target-level inventory. The saved lower-target directions do not establish robustness to an arbitrary tail level, actual shortage costs, or a new economic decision rule. Failed eligibility is a result of the specified procedure and remains part of the paper's boundary.

S23 Horizon coherence and late conditional-hit evidence

Realized peaks are ordered as the forecast horizon grows, but independently fitted reports use horizon-specific completed responses and can cross. The following exact strict comparisons keep all common issue origins. Repairing a long-horizon report upward can worsen pinball loss; target ordering alone does not identify a universally beneficial repair.

Table S24: Every original QR horizon-coherence comparison. Ordered targets do not force ordered fitted reports.

Source	W	Origins	Report crossings	Target violations
FX	720	6625	640	0
FX	2880	6625	140	0
Bills	52	277	91	0
Bills	260	277	80	0

The late fixed-terminal conditional-hit diagnostic uses the null $E[I_t \mid \mathcal{F}_t] \leq \alpha$, where $I_t = 1\{P_t > C_t\}$. Within each globally anchored maturity phase, a fixed upper-probability alternative supplies hit/no-hit factors with conditional expectation at most one under the stated null. The procedure mixes whole products over its fixed prior, then averages dependent phases. It does not multiply phases or select their maximum. The fixed-family threshold does not restore retrospective error control after viewed-sample diagnostic selection.

Table S25: Every threshold-crossing cell in the complete late diagnostic. A noncrossing elsewhere is not a calibration certificate.

Source	H/W	Hits/origins	Saved display value
FX	24/720	924/6625	8.499112682788E+7
Bills	6/52	84/277	1.310958812771E+8
Bills	6/260	54/277	2.204375927802E+3
Bills	24/52	132/277	3.814063640027E+4

The complete summary and phase tables, exact rational identities, all noncrossings and the interpretation boundaries remain in the e-value archive. This is a fixed-terminal diagnostic, not an anytime p value, optional-stopping guarantee, new estimator, or independent validation. Score, breach amount, frequency and conditional-hit evidence answer different questions.

S24 Numerical implementation and inference conventions

Completed labels and update clocks. At a scheduled refit κ, the latest completed response origin is $\kappa - H$. A history of W completed responses therefore uses $\{\kappa - H - W + 1, \ldots, \kappa - H\}$, with every target ending by that refit. Training transforms, scales, coefficients and historical pools are held until the next scheduled refit. The current observed position may still change at each forecast issue. Replacing that issue anchor by the refit anchor changes the forecasting procedure.

Regression and local quantiles. The regression objective uses a training-normalized headroom response, an unpenalized intercept and a ridge penalty on slopes. Projection is applied

when issuing the report and evaluating its validation score, not inside the fitted training objective. Local methods use training-only standardized predictors, fixed neighbour menus and duplicate-preserving order statistics. A selected whole-pool local quantile is an exact historical identity, not a failed-fit fallback. Scale-adjusted neighbours change responses as well as retrieval.

Numerical eligibility. The accepted solver records primal and dual residuals, objectives, feasibility and their gap. Ill-conditioned, nonfinite, unsupported-rank or failed candidate states remain failures under the specified whole-cell selection contract. Changing the ridge menu changes the procedure even when feature definitions stay fixed. A successful numerical certificate is separate from the financial accuracy, economic relevance and inferential validity of a forecast.

Working-reference inference. Saved unconditional loss differences use the recorded Bartlett-HAC convention; HLN and doubled-bandwidth sensitivities are labelled separately. Separate families and later reporting partitions do not erase the wider selection history. Coverage references distinguish full-grid Kupiec from the transition-conditional Christoffersen joint likelihood, which conditions on the initial hit. Small sparse grids and boundary likelihoods retain their original status. Neither a nonrejection nor a small working-reference p value proves conditional coverage or supplies selection-adjusted confirmation.

Common-pool construction distances. On one nonempty historical episode set, compare the historical headroom, translated-path and historical-peak reports using the same available order statistic. Each episode peak includes its starting observation. A translated path changes each level by the same anchor difference, so its absolute peak changes by at most that difference. Flooring is nonexpansive and commutes with the fixed order statistic. The headroom construction has the corresponding bound in the difference of anchor magnitudes. Thus

$$|C^{\mathrm{headroom}} - C^{\mathrm{peak}}| \le \max_i \big||N_i| - B\big|, \qquad |C^{\mathrm{translated}} - C^{\mathrm{peak}}| \le \max_i |N_i - x|.$$

For a proof of the order-statistic step, paired coordinatewise differences bounded by a radius imply that each empirical quantile lies below the other plus that radius; reverse the roles for the second inequality. These deterministic same-pool bounds do not order financial accuracy or coverage. Changing historical rows on one side invalidates the common-pool argument.

Complete implementation evidence. Exact menus, numerical tolerances, failures and diagnostic tables remain in EXTENDED RESULTS and the dated source archives. This supplement describes the existing procedures; its document builder imports no forecasting code and performs no new financial computation.

S25 Reproduction, availability and retained extended results

What is public. The repository provides editable sources, public aggregate results, original design records, computational code and numerical verification evidence. Old favourable and adverse outputs remain unchanged. The compact supplement links the longer retained material instead of removing its rows. Exact source-field bindings for every aggregate printed here accompany the document generator.

What public checking proves. The read-only conditional-archive entry point verifies the existing aggregate inventory without raw observations. Document regeneration reads public

aggregates and formats their saved values. These checks establish internal artifact consistency, not recomputed forecasting accuracy, a new sample, independent scientific replication or journal acceptance.

Full financial replay requires compatible inputs. The original FX member and Treasury snapshot must match their recorded fingerprints and source definitions before a separately authorized fresh replay. Private per-origin forecasts, fitted traces and restricted FX observations are not redistributed. Downloaded-vintage economic dates differ from verified historical first-release availability. A current file with the same series name is not automatically the accepted snapshot.

Version-specific entry points. The v7 merge preserves both source lines. The supervisor panel scripts retain their original canonical paths and whole-file identity gates. Earlier panel implementations remain as the explicitly named legacy scripts, with the same original bytes. Historical execution contracts keep their original values and dates; exact old-protocol replay should use its recorded source commit. The existing prospective parameter binding is not silently reactivated or rebound by this manuscript edit.

Resources and software. This revision reads existing local software and results. No package installation, model fit, additional data acquisition, simulation rerun or parameter search belongs to document assembly. The historical execution receipts report their actual worker time, resident memory and observed threads separately from requested budgets. Test execution uses an isolated copy of tracked inputs so synthetic fixture writes do not alter the scientific archives.

Routes to complete evidence. The v6 replay guide, original conditional archive, supervisor rule archive and extended-results index identify the complete retained tables, failure inventories and longer technical discussions. Missing compatible external validation and prospective first-release evaluation remain research tasks, rather than claims created by these reproducibility materials.